DunesForge - installation and updates

Version 0.85 beta | Keep the ZIP and backups of your projects.

New here? Start with Getting Started (ten minutes). Something not working? See Troubleshooting.

GPU repair and offline mode

The installer respects Blender's internet access setting. To download GPU libraries, open **Edit > Preferences > System > Network** and enable **Allow Online Access**. An already working GPU installation can also be checked offline. You can continue working on CPU without internet access.

If a previous installation was interrupted or its settings are damaged, choose **Install CuPy GPU Backend** again. The installer prepares a new set of libraries and activates it only after successful verification. It keeps a backup of the settings file it replaces. Press **Esc** to cancel downloading or GPU verification; you can retry after cancellation or failure. Restart Blender after a successful installation.

Step-by-step installation

1. Download the complete DunesForge ZIP file (for this beta, for example **DunesForge_0.85.5_beta.zip**). Do not extract it first.
2. Start Blender 5.1 or newer and open **Edit > Preferences**.
3. In the extensions section, open the top-right menu and choose **Install from Disk**. Select the ZIP. Blender also supports dropping an extension ZIP into its window.
4. Check that **DunesForge - Dune & Snow Simulator** is enabled in preferences.

5. Return to the 3D Viewport, press **N** and open **DunesForge**. Its build label should begin with **0.85**.

6. Click **QuickStart**. Choose PL or EN beside the help buttons. Manuals are stored in the installed extension's **docs** subfolder and work offline.

Local ZIP installation is also covered by the official Blender manual. The preference layout can vary slightly between Blender versions.

Optional GPU acceleration

The add-on also starts without GPU libraries. You can work immediately on CPU; begin with a small grid. CUDA acceleration needs a supported NVIDIA GPU and compatible driver. AMD/Intel GPUs and macOS use CPU.

1. Open DunesForge and read its CPU/GPU status.

2. On a supported NVIDIA machine click **Install CuPy GPU Backend**. Internet and free disk space are needed; the complete library set can take several GB.

3. Watch the stage message. Downloading runs in the background. **Escape** cancels downloading; do not start a second installation at the same time.

4. The installer selects CuPy compatible with Blender's bundled NumPy and downloads required CUDA components. You do not need a separate CUDA Toolkit or administrator access to Blender's folder.

5. Wait for the array, kernel compilation and FFT tests. Only a successful test activates the new installation.

6. Close and restart Blender. Confirm **GPU Acceleration: ACTIVE**, then try a small scene.

Libraries live under `~/.tydunes/gpu` in your home folder, separated by Python, operating system and NumPy version. On Windows `~` usually means `C:/Users/YOUR_NAME`. The installer does not

change Blender's NumPy or other bundled libraries. Each attempt creates a new directory, and an earlier working set stays active until the new set passes verification.

If installation fails, keep the message and the log path it reports. A missing NVIDIA driver must be installed or updated from the GPU manufacturer; DunesForge does not install system drivers. With no internet, a write permission problem or no compatible package, you can continue on CPU and retry after resolving the cause. After a Blender update, check the status: a different Python or NumPy version may need a new GPU set. Do not copy libraries between versions manually.

First check

Open a fresh scene, select **Desert** and click **Create Demo Scene**. Let it finish preparing. Inspect terrain, materials and a few playback steps. Start small. Before applying the setup to an important project, save a test and confirm you can reopen it.

Updating an existing installation

1. Save the project, external files and any simulation state you need. Keep the previous ZIP.
2. Stop playback. Install the new local ZIP through **Install from Disk**, using the update/replace option available in your Blender version.
3. Close and restart Blender. Replacing files does not remove already loaded Python code from memory.
4. Check the build label and open a copy of the project. Validate it briefly before returning to the main version.

Finding the manuals

The **docs** folder contains **GettingStarted_PL/EN** (start here), **Troubleshooting_PL/EN** (common problems), **QuickStart_PL/EN**, **Reference_PL/EN**, **Installation_PL/EN**, **Product_PL/EN** and **Showcase_PL/EN**. All of them are **PDF** files, in Polish and in English, and work offline. Use your PDF viewer's search (Ctrl+F) to find a word.

The add-on buttons open them in your PDF viewer. Panel help opens the corresponding chapter; right-click help opens the exact setting. Opening the exact place works with Adobe Acrobat or Reader, SumatraPDF and the PDF viewers of Edge, Chrome, Firefox, Opera, Brave and Vivaldi; other viewers open the guide at its first page. If a manual is reported missing, reinstall the complete ZIP rather than copying only code files.

Common problems

- **No DunesForge tab:** confirm the add-on is enabled and you are using the 3D Viewport sidebar.
- **Old build label:** restart Blender and check that you updated the Blender installation you are currently running.
- **GPU unavailable:** check driver and dependency compatibility; use a smaller CPU test meanwhile.
- **Missing textures:** install the complete package and check asset paths, especially after moving a project.
- **Enable error:** copy the complete console message together with Blender and add-on versions.

The supported minimum is Blender 5.1. The add-on still installs on Blender 4.2 to 5.0 and may work there, but those versions have not been thoroughly tested. Tests do not mean every feature has been checked on every Blender and hardware combination.